**Foundations Practice Exam**

**Question 1** **[2+2+2+2=8 marks]**

Write your responses to the following in the boxes:

(a) The slope of the position-time graph of a particle's motion gives which of the following:

1. The particle's displacement;
2. The particle's velocity;
3. The particle's average acceleration;
4. The particle's instantaneous acceleration?

(b) A student pushes a very heavy box across a horizontal floor. The work done by gravity on the box:

1. Depends of the weight of the box;
2. Cannot be calculated without more information;
3. Is equal to zero;
4. Depends on the friction forces involved.

(c) Complete the following sentence with the **one** correct response.
The impulse delivered to a body by a net force is:

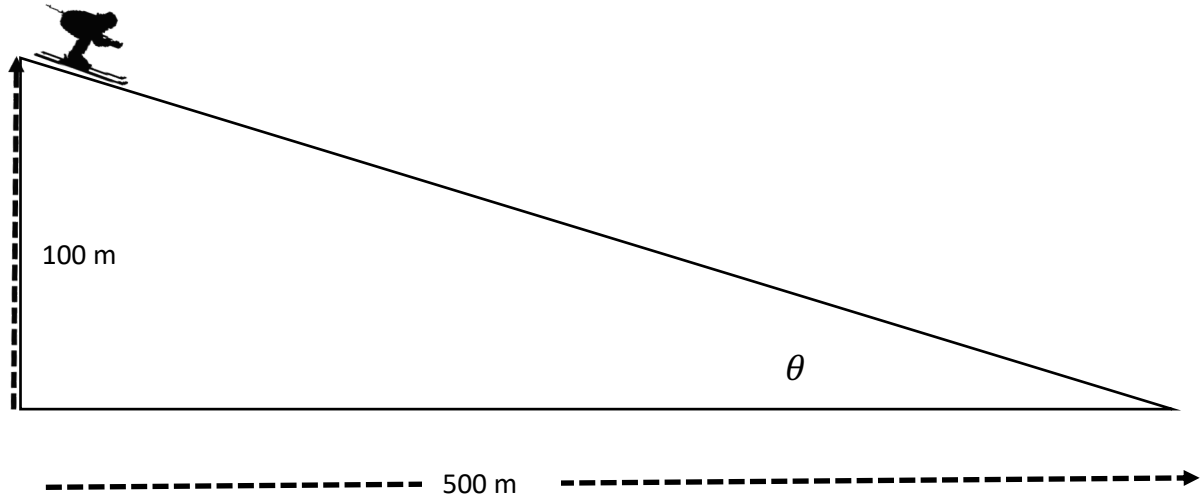
1. Defined only for interactions of short duration;
2. Equal to the change of momentum of the body;
3. Equal to the area under a force versus position graph;
4. Defined only for elastic collisions.

(d) The sound intensity for the whistle of an approaching train is recorded as 60 dB. Select the **one most correct** statement from the following.

1. Twice the sound intensity would be 120 dB.
2. The sound intensity when the train is at half the distance would be 120 dB.
3. If the train is at one third the original distance, the sound intensity would be about 70 dB.
4. If the sound intensity is 70 dB, the train is at one tenth the original distance.

**Question 2** **[4+2+4+4+6=20 marks]**

A skier stands at the top of a slope 100 m high and 500 m long as shown in the figure below. In the calculations that follow, neglect the effects of friction. We intend to determine the skier's acceleration down the slope and the time taken to reach the bottom of the slope. Assume the acceleration due to gravity $g = 9.8\ \mathrm{ms}^{-2}$.

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(a) Draw a free body diagram in the box on the next page showing all the forces acting on the skier. Define an appropriate coordinate system in $x$ and $y$ to perform the calculations and draw them near your free body diagram.

(b) Determine the angle $\theta$ to the horizontal of the slope.

(c) Resolve the forces acting on the skier into components parallel and perpendicular the slope.

(d) Apply Newton's Laws of motion to determine the skier's acceleration.

(e) Calculate the time for the skier to reach the bottom of the slope.

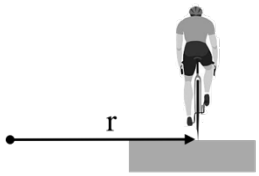
**Please answer all questions on next page**

Question 2 a)-e)

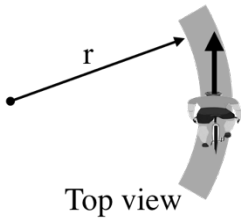
**Question 3** **[3+3+2=8 marks]**

A lecturer-cyclist does a left turn at constant speed without slipping on a horizontal road by following a circular path of radius 25 m.

a) Draw a clearly labelled free-body diagram showing all forces acting on the bicycle as it follows the circular path. Draw the diagram using the rear view of the bicycle.
b) If the coefficient of static friction is $\mu_s = 0.7$, how fast can the bicycle travel without slipping?
c) Explain how the cyclist can be accelerating while travelling at a constant velocity? Include an arrow representing the acceleration of the cyclist on the Top View diagram




Rear view




Top view

**Question 4** **[5 marks]**

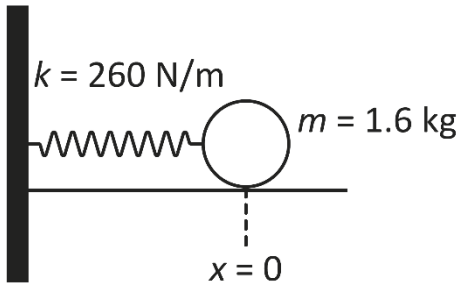
Two aspiring astronauts, Frida and Ahmad are training for when they will be on the International space Station in orbit around the Earth. They are discussing being weightless while being on the space station.

Frida insists that they will feel weightless because they are out of the Earth's gravitational field. Ahmad disagrees and thinks they will feel weightless because they are falling.

Who is correct? Explain in terms of gravitational orbits why they are correct. You may use diagrams to support your answer.

**Question 5** **[2+4+2=8 marks]**

A clockmaker is fabricating a new instrument to measure time. Initially, the design of the instrument is a spring connected to a mass as shown.




The mass is initially displaced so as to compress the spring, storing $250\ \mathrm{mJ}$ of energy. The mass is then released and undergoes simple harmonic motion.

(a) Write an expression for the displacement of the mass as a function of time.

(b) Using your answer for part (a), derive an expression for the acceleration of the mass as a function of time and hence calculate the magnitude of the maximum acceleration experienced by the mass.

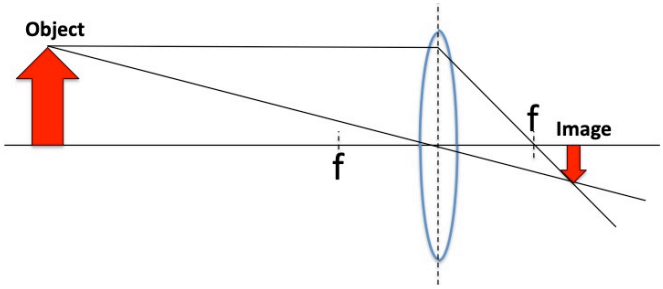
(c) The clockmaker observes significant friction present in the system. *Briefly* (1 or 2 sentences) describe how the motion of the mass will be altered.

**Question 6** **[1+3+1+2=7 marks]**

(a) A person has eyeglasses with positive (converging) lenses. Are these lenses most likely to assist with:

(i) Far vision; i.e. clearly seeing objects that are far away
(ii) Near vision; i.e. clearly seeing objects that are near
(iii) Both near and far vision, because the person has presbyopia.
(iv) Cannot tell from information given.

The diagram shows the object and image typical of a camera.



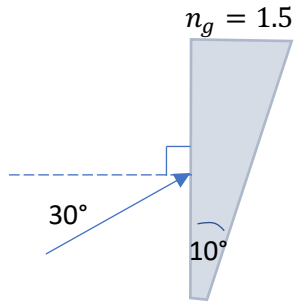

b) How far from the camera lens must the detector be, if the lens focal length is 50 mm, to focus on an object 75 cm away?

c) Is the image real or virtual?

d) What is the magnification of the image?

**Question 7** **[2+3+3=8 marks]**

A laser beam propagating in air (refractive index $n_a = 1.0$) is incident on a glass wedge (refractive index $n_g = 1.5$) at angle 30° (see figure). The two surfaces of the glass wedge are at an angle of 10° to each other. Calculate




a. The angle of the light after the first surface (relative to normal)

b. The angle of the light exiting the glass, relative to the incident ray.

c. Is it possible to change the incident angle such that the beam is totally internally reflected from the second surface? Justify your answer.

**Question 8** **[3+2=5 marks]**

A 5.0 kg block with a speed of 3.0 m/s collides with a 10.0 kg block that has a speed of 2.0 m/s in the same direction. After the collision, the 10.0 kg block travels in the original direction with a speed of 2.5 m/s.

a) What is the velocity of the 5.0 kg block immediately after the collision?
b) Determine the magnitude of the change, if any, in the total kinetic energy of the system of two blocks because of the collision.
c) Is the collision elastic or inelastic

**Question 9** **[3+3=6 marks]**

Usain Bolt, the current 100 m world-record holder, starts a race from rest and travels in a straight line under constant acceleration to reach 43.2 km/hour after 2.9 seconds. For the rest of the race he travels at constant velocity until he passes the finishing line.

a. How long did he take to run the 100m?

In his next 100m race, Usain Bolt runs at a constant velocity of 40.7 km/hour until he passes the finishing line.

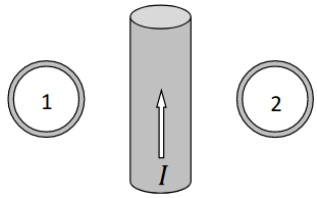
After crossing the line, he undergoes constant deacceleration to slow to a stop in 4.0 seconds.

b. Calculate his total displacement from the start of the race until he stops.

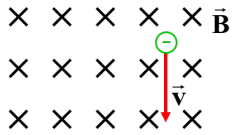
c.

**Question 10** **[2+1+3+3+4=13 marks]**

a) The figure below shows two closed loops of wire, 1 and 2, near a current-carrying wire *I*. If that current is **decreasing** with time, what direction will the induced currents in the two loops have?




b) The figure below shows a negatively charged particle moving downwards at some velocity *v*, within a uniform magnetic field directed into the page. Please draw the direction of the force on the particle in or below the figure:

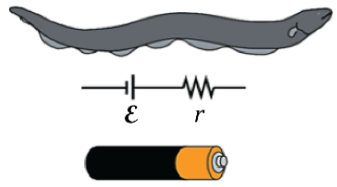


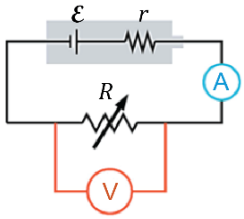

c) In a lecture demonstration a capacitor is charged and then electrically isolated.

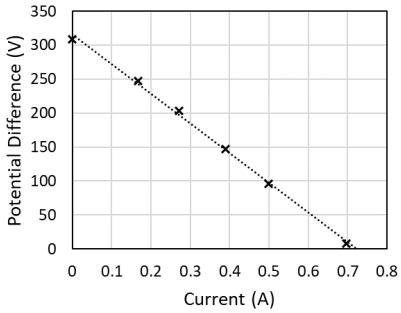
1. How does the energy stored in the system change as the separation between the plates is increased? Explain.
2. If, on the other hand, a battery is connected across the plates, how does the stored energy change as the separation between the plates is increased? Explain.
3. If the capacitor is charged and kept electrically isolated and a sheet of glass with a dielectric constant $\kappa$ is introduced into the region between the plates, how does the stored energy compare with that for an identical capacitor with vacuum between the plates? Explain.

**Question 11** **[2+1+3+4+2+2+2+2+2=20 marks]**

Electric eels will undertake a 'leaping attack' emerging from water to deliver an electric shock to a threat. This defense strategy has been used against humans. In this question we model the resulting electric shock using an equivalent electric circuit. The circuit models the eel as a battery with electromotive force (emf) $\boldsymbol{\varepsilon}$ and internal resistance $\boldsymbol{r}$.




To measure the eel emf and resistance, a scientist undertakes a series of measurements of the current through the eel and the associated voltage across a variable resistor $\boldsymbol{R}$ as this resistance is varied. Note that the symbol in the figure above right refers to a variable resistor where the resistance can be changed.

The results of the measurements obtained from one eel are shown in the graph above along with a straight line of best fit.

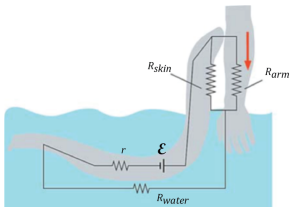
Based on the information in this graph

(a) Show that the potential difference, $\boldsymbol{V}$, of the variable resistor can be represented by
$\boldsymbol{V = \mathcal{E} - Ir}$.

(b) Write down an estimate of the value of the emf $\boldsymbol{\mathcal{E}}$ of the eel.

(c) Write down an estimate of the value of the internal resistance of the eel $\boldsymbol{r}$.

If the eel touches the arm of a person whose hand is submerged in water, some of the current will pass through the length of arm between the point of contact and the surface of the water and the remainder along the surface of the eel's skin as shown in the figure. The return path to the tail of the eel is through the water. We can model this with a circuit as shown in the figure below right with a resistance $\boldsymbol{R_a}$ in parallel with the path along the eel's body $\boldsymbol{R_s}$ and the return path through the water with resistance $\boldsymbol{R_w}$. In what follows take the resistance of the lower half of the human arm to be $\boldsymbol{R_a \approx 2,100\ \Omega}$, the eel's skin $\boldsymbol{R_s \approx 2,400\ \Omega}$ and the water to have a resistance $\boldsymbol{R_w = 400\ \Omega}$.




(d) Show that the equivalent resistance of this circuit is 1,950 Ω ?

(e) How much current flows through the water?

(f) If the potential difference across the arm is 187 V, calculate the current that flows through the arm?

(g) Would the person be able to perceive this current? Explain

(h) How much power is delivered by the eel emf?

(i) How much power is dissipated in the arm?

**Please write all answers to this question on the next page**

Answers Question 11 a)-i)

**Question 12** **[2+2+2+3=9 marks]**

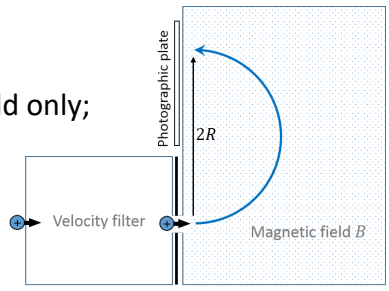
A Wien ion velocity filter, shown in the diagram below, uses crossed uniform electric and magnetic fields to select charged particles according to their velocity.

(i) Sketch the arrangement of electric and magnetic fields used in the velocity filter.

(ii) Let $v$ be the velocity of the particles which pass through the filter. Show on your sketch charged particle trajectories for particles with velocity $v_{\text{low}} < v$ and $v_{\text{high}} > v$, labelling each trajectory accordingly.

(iii) For constant electric and magnetic field strengths $E$ and $B$, and assuming that all the particles have the same charge, explain why only particles whose velocity is $v = E/B$ pass through the filter.

The filter can be followed by a region of uniform magnetic field only; that is, with no electric field, to create a type of mass spectrometer, as shown in the diagram below.




iv) Derive an expression for the mass $m$ of each charged particle in terms of q, B, R and v

**Question 13** **[4 marks]**

Transcranial magnetic stimulation (TMS) is a non-invasive procedure that uses magnetic fields to stimulate nerve cells in the brain to improve symptoms of depression. Briefly describe how this technique can be used to excite neurons up to 5cm beneath the skull.

**Question 14** **[3+2+2+2+3+2=14 marks]**

The nuclear reactor at the Australian Nuclear Science and Technology Organization (ANSTO) creates the isotope Mo-99 which then decays to the metastable isotope Tc-99m. Tc-99m is used as a radioactive tracer for helping to determine the cause of a patient's illness. The Tc-99m tracer emits a gamma ray during its decay to Tc-99. Tc-99 then further decays to Ru-99 which is a stable isotope.

The decay series and relevant information for each isotope is given in the table below;

| **Name** | | **Isotope** | **Mass (u)** | **Half-life** | **Decay** |
|---|---|---|---|---|---|
| Molybdenum | Mo-99 | ${}^{99}_{42}Mo$ | 98.9077119 | 2.7474 days | Beta |
| Metastable Technetium | Tc-99m | ${}^{99m}_{43}Tc$ | 98.9064065 | 6.0058 hrs | Gamma |
| Technetium | Tc-99 | ${}^{99}_{43}Tc$ | 98.9062547 | 211,000 yrs | Beta |
| Ruthenium | Ru-99 | ${}^{99}_{44}Ru$ | 98.9059393 | stable | |

Rest masses of: ${}^{4}_{2}He$ atom: 4.002602 u; electron: 0.000549 u; positron: 0.000549 u

${}^{1}_{1}H$ atom: 1.007825 u; neutron: 1.008665 u; proton: 1.007276 u

a) The Molybdenum is transported to hospitals and produces the metastable Technetium to be injected into the patient. Mo-99 decays to the metastable Tc-99m via beta decay according to the equation:

$$ {}^{99}_{42}Mo \rightarrow {}^{99m}_{43}Tc + {}^{0}_{-1}\beta \quad + \bar{\nu} + energy $$

i. Show that the energy released during this decay is 1.216 MeV.

ii. Where does the energy released go?

iii. Tc-99m then undergoes radioactive decay to Tc-99 by releasing a 141 keV gamma ray as it moves from its excited state to the ground state. What is the wavelength of the resulting gamma ray released?

iv. What would you advise regarding the transportation of the radioactive materials to protect the person handling the materials?

(b) Patients are injected with the metastable Technetium which has been separated from the Molybdenum. The resulting gamma rays emitted as the isotope decays are an excellent radioactive source for imaging.

I. Calculate the percentage of Tc-99m that will still be present in the patient's body after 1 day.

II. Both Mo-99 and Tc-99 are beta sources, yet only Tc-99 remains in the patient's body as Mo-99 is removed from the sample before the procedure. Using the data in the table, compare the amount of radiation received by each of the sources and comment on why it is fine for Tc-99 to remain in the body, but not Mo-99. Include calculations to justify your answer

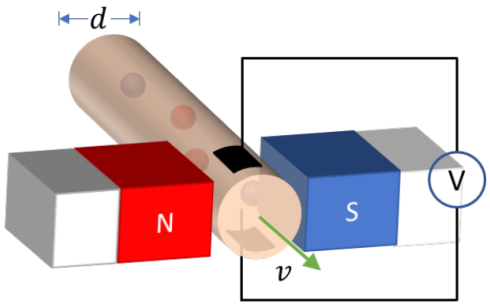
**Please answer all question on next page**

Answers Question 14 a i)-iv) and 11b i), ii)

**Question 15** **[1+2+3+2=8 marks]**

The diagram shows positive (Na+ ) and negative (Cl- ) ions in the blood flowing in an artery at a speed v as shown.
A magnetic field of strength B=0.05T is applied across the artery which has a diameter d=4mm.

Na+ is an ion with one electron missing, and Cl- is an ion with one extra electron.




a) Write down an expression for the magnitude of the magnetic force on the ions in terms of quantities provided above.

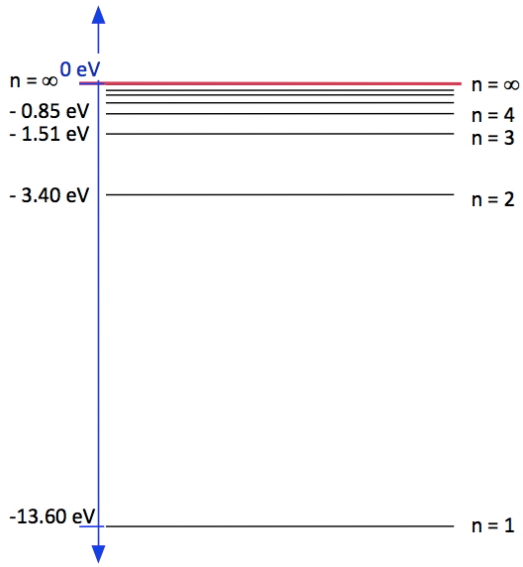
b) What will be the direction of the magnetic force on the
   a. positive ions and
   b. on the negative ions?

c) We have a potential difference of 80.0 μV across the artery, what is the speed of the blood?

d) Explain why an electric field appears across the artery.

**Question 16** **[4+3=7 marks]**

The diagram shows the energy of the ground state of the hydrogen atom, as well as the energies of three of its excited states.




1. A hydrogen atom absorbs radiation so that it is excited to the third energy level above the ground state. What are all of the possible energies, in eV, of photons that could then be emitted by the hydrogen atom?
2. Suppose in a sample of hydrogen atoms at room temperature the atoms are all in the ground state. Justify the statement that none of the radiation absorbed by any of these hydrogen atoms in the ground state is in the visible part of the electromagnetic spectrum.